

بسم الله الرحمن الرحيم

Kingdom of Saudi Arabia
Ministry of Higher Education
College of Computer
Computer Science Department



المملكة العربية السعودية
وزارة التعليم العالي
كلية الحاسب
قسم علوم الحاسب

MATH317 : Numerical Methods

MedTerm Exam Mon. 10/10/2022

الرقم الجامعي

اسم الطالب :

Question1: Answer the following:

(2 + 3 m)

1. What is the relative error involved in rounding 3.9997 to 4.000?

2. Deflate the polynomial P , using the fact that 2 is one of its zeros, where P is the polynomial:

$$P(x) = x^4 - 4x^3 + 7x^2 - 5x - 2$$

Question2: Given the following function:

(6 m)

$$f(x) = x^3 - 3x^2 + 8x - 5$$

1. Write the Taylor series for the function $f(x)$, using $x = 3$ as the point of expansion.
2. Compute the approximate value of $f(3.02)$

Question 3:

(7 m)

Determine the single-precision and double-precision machine representation of the decimal number: 94.78125.

Question 4: Given the following function:

(7 m)

$$f(x) = 9x^3 + 6x^2 - 12x - 2$$

Find a root of the $f(x)$ in the interval $[0, 1]$ by using the bisection method: Terminate the iterations if they exceed $n_{\max} = 3$ or if the error bound is less than 0.5×10^{-6} .